Draft version November 10, 2025

Typeset using LATEX **twocolumn** style in AASTeX631

PageRank Analysis Using Iterative and Eigenvalue Methods on Directed Web Graphs

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ABSTRACT

abstract

Keywords: stochastic matrix — hyperlink — directed graph — word4 — word5

1. INTRODUCTION

NOTE: re-write/reorganize intro, data/observations, and results section for better flow/cohesiveness

The PageRank algorithm, first introduced by Larry Page and Sergey Brin, was one of the first algorithms to rank webpages based on their link structure rather than solely content Brin & Page (1998). The algorithm models the web as a directed graph and simulates a "random surfer" who follows hyperlinks with some probability. Essentially, it tracks how often this "random surfer" would land on a particular page over an extended period, so pages that are visited more frequently are considered more important and thus receive a higher PageRank score Page et al. (1999).

This setup defines a Markov process where each page's importance is proportional to the importance of pages linking to it. The final PageRank vector corresponds to the steady-state distribution of this process and can be found either by iteration or by solving for the dominant eigenvector of the hyperlink matrix $\boldsymbol{H}$ Langville & Meyer (2009).

In this project, we first implemented PageRank using the difference equation $\boldsymbol{x}_{k+1} = \boldsymbol{H}x_k$, which resulted in the scores converging once the solution reached the maximum tolerance allowed. We then confirmed our result by solving for the eigenvector associated with the largest eigenvalue of the matrix $\boldsymbol{H}$. Our goal is to understand how the algorithm behaves under different network structures and how quickly it converges Langville & Meyer (2004).

2. DATA AND OBSERVATIONS

For an $n \times n$ hyperlink matrix $\boldsymbol{H}$ to be stochastic, each column msut be a probability vector with non-negative entries that sum to 1. This ensures that probability is conserved at each step of the Markov process and allows the system to converge to a unique steady-state solution. This condition is only satisfied if every page has at least one outgoing link. Otherwise, the corresponding column in $\boldsymbol{H}$ would contain all zeroes and violate the stochastic property.

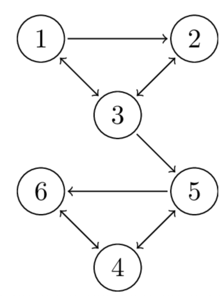

Figure 1. example graph 1.

To start, we defined a 6-node directed graph where each page links to one or more pages, as shown in Figure 1. From this, we constructed the hyperlink matrix $\boldsymbol{H}$, where each non-zero entry $\boldsymbol{H}_{ij}$ is equal to $\frac{1}{|P_{ij}}$, the reciprocal of the number of outbound links from page P_j. This matrix represents the probability that a random surfer on page j clicks through to page i:

$$\boldsymbol{H} = \begin{pmatrix} 0 & 0 & 1/3 & 0 & 0 & 0 \\ 1/2 & 0 & 1/3 & 0 & 0 & 0 \\ 1/2 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1/2 & 1 \\ 0 & 0 & 1/3 & 1/2 & 0 & 0 \\ 0 & 0 & 0 & 1/2 & 1/2 & 0 \end{pmatrix} \tag{1}$$

We initialized the PageRank vector $\boldsymbol{x}_0$ with equal values and assumed that each page is equally likely to be visited at first:

$$\boldsymbol{x}_0 = \begin{pmatrix} 1/6 \\ 1/6 \\ 1/6 \\ 1/6 \\ 1/6 \\ 1/6 \end{pmatrix} \tag{2}$$

We then applied the iterative update rule $\boldsymbol{x}_{n+1} = \boldsymbol{H}\boldsymbol{x}_n$ until the rank vector converged below a given tolerance to simulate a random surfer moving moving through the web graph via the links. We visualized the convergence of the PageRank vector by plotting each page's rank score for each iteration, which allowed us to track how the scores changed until they reached a steady state. Our results are shown in Figure 2.

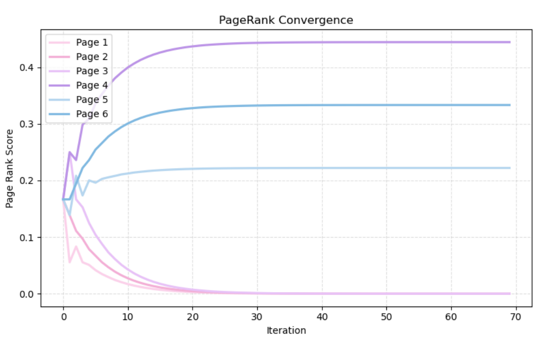

Figure 2. PageRank convergence for graph 1.

The convergence of the iterative method is a direct consequence of the underlying theory of stochastic matrices. According to Theorem 4.9, if $\boldsymbol{H}$ is an $n \times n$ stochastic matrix and $\boldsymbol{x}_0$ is some initial vector, then the system defined by the difference equation $\boldsymbol{x}_{n+1} = \boldsymbol{H}\boldsymbol{x}_n$ converges to a unique steady-state vector. This limit corrresponds to the eigenvector $\boldsymbol{v}_1$ associated with eigenvalue $\lambda = 1$, since all the other eigenvalue components vanish as $k \to \infty$:

$$\boldsymbol{x}_{\text{equilibrium}} = \lim_{k\to\infty} \boldsymbol{H}^k \boldsymbol{x}_0 = c_1 \boldsymbol{v}_1 \tag{3}$$

Theorem 4.9 greatly simplifies the PageRank iterative process. Instead of repeatedly applying $\boldsymbol{x}_{n+1} = \boldsymbol{H}\boldsymbol{x}_n$, we can directly compute the dominant eigenvector of $\boldsymbol{H}$ corresponding to the eigenvalue $\lambda = 1$. This vector represents the steady-state solution and results in the same PageRank distribution as the limit of the iterative method.

Both the iterative difference equation and the eigenvector method yield the same steady-state PageRank vector:

$$\boldsymbol{x}_{\text{converged}} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 4/9 \\ 2/9 \\ 1/3 \end{pmatrix} \tag{4}$$

This distribution reflects the long-term behavior of a random surfer nagivating the web graph. Based on the values, we rank the pages in order of importance as follows:

Page 4 > Page 6 > Page 5 > Page 1 = Page 2 = Page 3

This confirms that Pages 4–6 FINISH WRITING INTERPRETATION OF FINAL VEC HERE

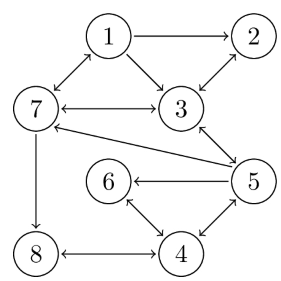

Figure 3. example graph 2.

We applied the PageRank algorithm to a second web graph consisting of eight interconnected pages, as shown in Figure 3. The corresponding hyperlink matrix $\boldsymbol{H}$ and initial uniform state vector $\boldsymbol{H}$ are defined below.

$$\boldsymbol{H} = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 1/3 & 0 \\ 1/3 & 0 & 1/3 & 0 & 0 & 0 & 0 & 0 \\ 1/3 & 1 & 0 & 0 & 1/4 & 0 & 1/3 & 0 \\ 0 & 0 & 0 & 0 & 1/4 & 1 & 0 & 1 \\ 0 & 0 & 1/3 & 1/3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/3 & 1/4 & 0 & 0 & 0 \\ 1/3 & 0 & 1/3 & 0 & 1/4 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/3 & 0 & 0 & 1/3 & 0 \end{pmatrix} \tag{5}$$

$$\boldsymbol{x}_0 = \begin{pmatrix} 1/8 \\ 1/8 \\ 1/8 \\ 1/8 \\ 1/8 \\ 1/8 \\ 1/8 \\ 1/8 \end{pmatrix} \tag{6}$$

Using both the iterative update method and the dominant eigenvector approach, we computed the steady-state PageRank distribution. The convergence behavior of the iterative method is shown in Figure 4

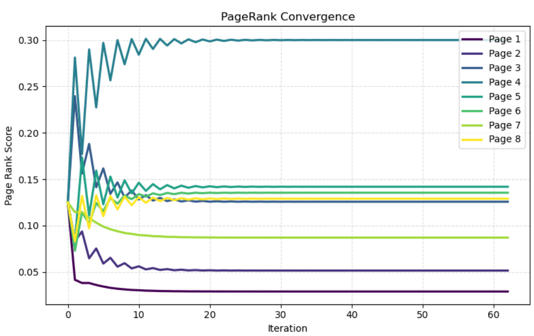

Figure 4. PageRank Convergence for graph 2.

Using both the iterative difference equation and the dominant eigenvector method, we arrive at the same steady-state PageRank vector:

$$\boldsymbol{x}_{\text{converged}} = \begin{pmatrix} 0.0290 \\ 0.0516 \\ 0.1258 \\ 0.3000 \\ 0.1419 \\ 0.1355 \\ 0.0871 \\ 0.1290 \end{pmatrix} \tag{7}$$

From this result, we rank the pages in order of importance:

$$\text{Page 4} > \text{Page 5} > \text{Page 6} > \text{Page 8} > \text{Page 3}$$
$$> \text{Page 7} > \text{Page 2} > \text{Page 1}$$

Page 4 is the most dominant, whereas Page 1 has the lowest influence in the network. (ADD SMTH FOR VECTOR INTERPRETATION HERE)

To summarize, we implemented two numerical approaches to compute the PageRank vector. The first was an iterative method that started from a uniform initial distribution $\boldsymbol{x}_0$ and repeatedly applied the update equation $\boldsymbol{x}_{n+1} = \boldsymbol{H}\boldsymbol{x}_n$ until the difference between successive iterations fell below a convergence threshold. The second approach used Theorem 4.9, which states that the steady-state distribution is the dominant eigenvector of the stochastic matrix $\boldsymbol{H}$. Both methods gave identical results, which confirms the accuracy of our implementation. We performed all computations in Python using `NumPy` and visualized convergence behavior with `matplotlib`.

3. RESULTS

NOTE: results n stuff from data section to here (I forgot this was a whole section itself)

4. SUMMARY AND CONCLUSION

In this project, we implemented the PageRank algorithm on two web graphs using both an iterative difference equation and the dominant eigenvector method. Our results agreed in both cases, which confirms the theoretical convergence properties of stochastic matrices....

4.1. *Acknowledgements*

We thank Professor Mitra for his guidance and instruction throughout the course, as well as the teaching assistants for their helpful feedback. We also acknowledge the Department of Physics for providing an excellent academic foundation to support this work. This work was completed as part of the requirements for C S 323E: Elements of Scientific Computing at the University of Texas at Austin.

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